

Unit	Details	No. Hrs
1	Concept of Manufacturing- Manufacturing definition; Role of materials, processes and systems in manufacturing; Classification and brief introduction of engineering materials such as metals & alloys, Classification and brief introduction of manufacturing processes	4
2	Sand Casting Process of Metals- Elements of Green Sand Mould; Pattern design and making, Method of Preparation of Green Sand Mould; Casting Defects	2
3	Metalworking Processes- Classification of Metalworking Processes-brief introduction of bulk and sheet metal processes, Hot Vs Cold Working; Hot and Cold Rolling; Types of Rolling Mills, Forging, Extrusion, Drawing	3
4	Machining Processes: Classification of machining processes & machine tools; Construction, Specification, and operations on Lathe Machine and Drilling machine	2
5	Fabrication Processes- Classification of Welding Operations, Types of Joints & Welding Positions; Brief description of Arc, Resistance and Gas welding techniques. Brazing and Soldering	3

List of Practical

1. Safety in Workshop (Demonstration)

Safety precautions and utilization of hand tools and machines of different shops with safe working habits. Introduction to measuring equipments and gauges of different shops.

2. Carpentry

Study of wood works, types of hand tools and machine. Making of one job involving wood work joint

3. **Fitting**

Study of different fits and hand tools. Making of one job involving fitting to size, male-female fitting with drilling and tapping

4. **Welding**

Study of electric arc welding and gas welding, tools, types of weld joints and safety precaution during welding. Making of one joint using electric and gas welding. Students will be introduced to brazing and soldering (demonstration)

5. **Sheet Metal Work**

Study of different hand tools, machine and sheet metal joints. Making of one utility job in sheet metal

6. **Foundry**

Principles of molding, methods, core & core boxes, preparation of sand mould of given pattern and casting (demonstration)

7. **Black Smithy**

Introduction to hot working and Study of forging hand tools, furnace and machine. Making a job on hot upset forging.

8. **Machining**

Study of lathe machine, cutting tools and turning related operations. Making of one job on lathe machine including facing, step and taper turning, threading operations.

9. **Plastic Processing**

Introduction to plastics and different plastic molding techniques. Study of injection molding process with demonstration.

10. **Computer Numerical Control (CNC)**

Introduction to automation & CNC, Assembly of models of CNC, CNC wood router, engraving and exposure to part programming. Preparation of part program for simple profiles. Making a job on CNC (Demonstration).

11. **Mini Project**

Team activity – Fabrication of prototype model based on above practical.